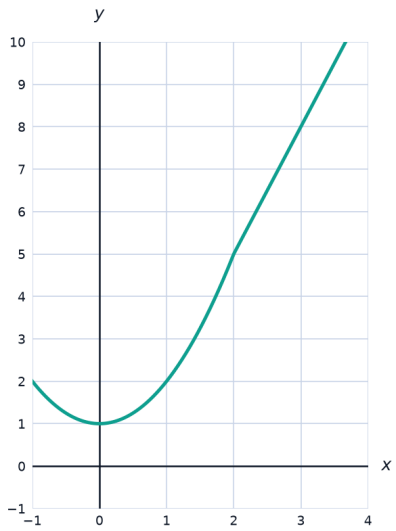
Continuity

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Testing continuity**

Consider the piecewise function $f(x) = x^2 + 1$ for $x < 2$, and $f(x) = 3x - 1$ for $x \geq 2$.



a Find $\lim_{x \to 2^-} f(x)$.

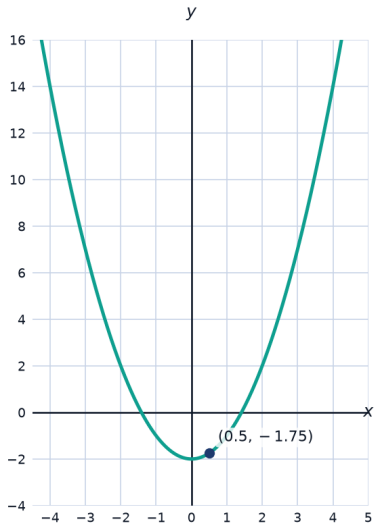
b Find $\lim_{x \to 2^+} f(x)$.

c Is f continuous at $x = 2$? Justify.

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2 [Foundational] • Target: Testing continuity

The piecewise function $f(x) = kx + 1$ for $x \leq 3$, and $f(x) = x^2 - 2$ for $x > 3$, is continuous at $x = 3$. Find the value of k.



3 [Foundational] • Target: Types of discontinuity and the Intermediate Value Theorem

Classify the discontinuity of each function at the given point:

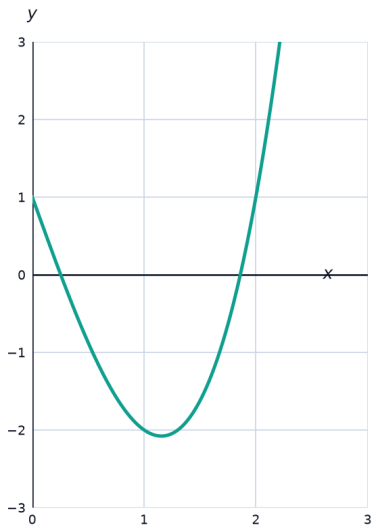
a $f(x) = \frac{x^2 - 1}{x - 1}$ at $x = 1$

b $f(x) = \frac{1}{x - 2}$ at $x = 2$

c $f(x) = \lfloor x \rfloor$ at $x = 3$

4 [Foundational] • Target: Types of discontinuity and the Intermediate Value Theorem

Show that $f(x) = x^3 - 4x + 1$ has a root in the interval $(1, 2)$, using the Intermediate Value Theorem.



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5 [Foundational] • Target: Continuity on an interval

State the largest interval(s) on which each function is continuous:

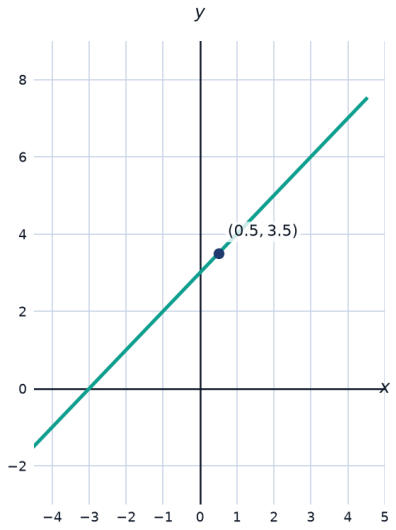
a $f(x) = \frac{1}{x-4}$

b $f(x) = \sqrt{x-3}$

c $f(x) = \ln(x+2)$

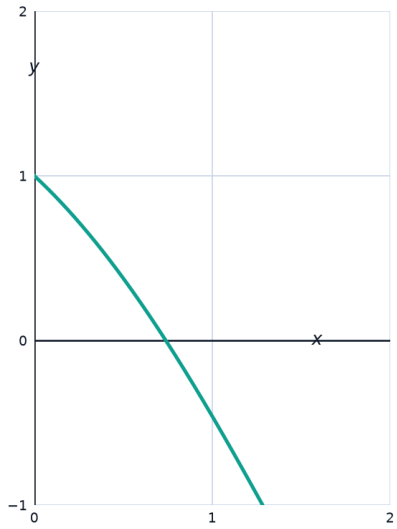
6 [Foundational] • Target: Continuity on an interval

Find the value of c that makes $f(x) = \frac{x^2-9}{x-3}$ for $x \neq 3$, and $f(3) = c$, continuous at $x = 3$.



7 [Foundational] • Target: Applying the Intermediate Value Theorem

Show that $f(x) = \cos x - x$ has a root in the interval $(0, 1)$.



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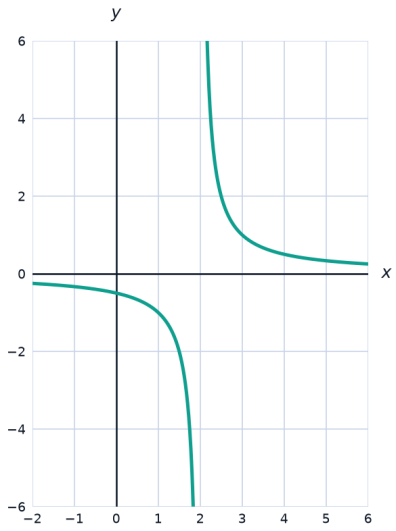
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Applying the Intermediate Value Theorem**

A hiker starts a trail at elevation 200 m at 8:00 AM and finishes at elevation 800 m at noon. Explain, using the IVT, why the hiker must have been at elevation exactly 500 m at some point during the hike.

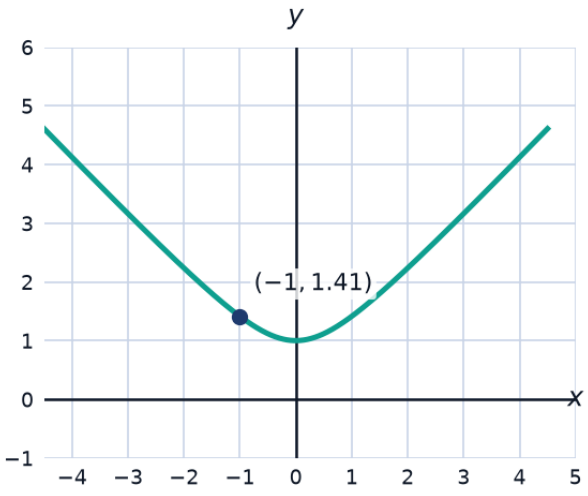
9 **[Intermediate]** • **Target: Visualizing discontinuities**

The function $f(x) = \frac{1}{x-2}$ (with an infinite discontinuity at $x = 2$) is graphed below. Use the graph to describe the one-sided behavior near $x = 2$.



10 **[Intermediate]** • **Target: Continuity of composite and combined functions**

Explain why the composition of two continuous functions is continuous, and use this to argue that $h(x) = \sqrt{x^2 + 1}$ is continuous everywhere.



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11 [Intermediate] • Target: Removable discontinuities and redefinition

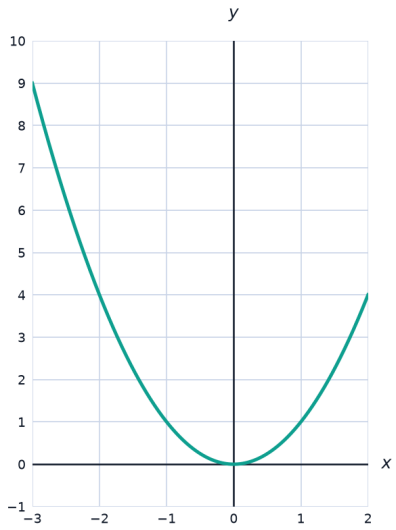
The function $f(x) = \frac{x^2 + x - 6}{x - 2}$ has a removable discontinuity at $x = 2$. Find the value that, if assigned to $f(2)$, would make f continuous there.

12 [Intermediate] • Target: Removable discontinuities and redefinition

Find all values of x where $f(x) = \frac{x^2 - 16}{x^2 - 2x - 8}$ has a discontinuity, and classify each as removable or non-removable.

13 [Intermediate] • Target: The Extreme Value Theorem (a continuity consequence)

State the Extreme Value Theorem, and explain why $f(x) = x^2$ is guaranteed to have an absolute maximum and minimum on $[-3, 2]$.



14 [Intermediate] • Target: Continuity of composite and combined functions

Level 3 · Advanced Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle constraint $g(x, y) = x^2 + y^2 = 1$ using the Method of Lagrange Multipliers.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

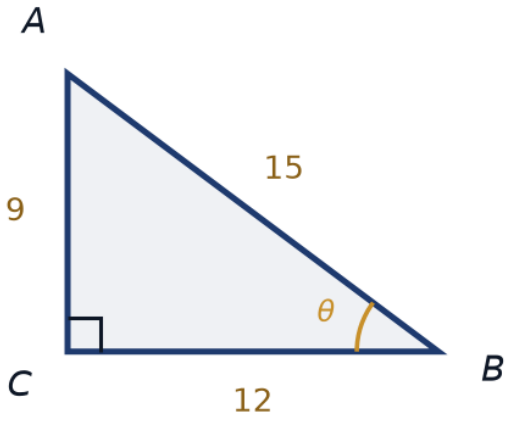
15 [Challenging] • Target: Removable discontinuities and redefinition • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Level 2 · Medium Use Green's Theorem to evaluate $\oint_C (3y\,dx + 5x\,dy)$ where C is the counterclockwise circle $x^2 + y^2 = 4$.

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16 [Challenging] • Target: The Extreme Value Theorem (a continuity consequence) • Cross-Lesson: 11.01 Sequences and Series

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ for the function: $y = \int_{\cos x}^{x^3} \sqrt{1 - t^2}\, dt$.



17 [Challenging] • Target: Testing continuity • Cross-Lesson: 12.01 Convergence Tests

Level 1 · FoundationsFind $h'(2)$ given the accumulation function $h(x) = \int_{0}^{x^2} \frac{t}{t+1}\, dt$.

18 [Challenging] • Target: Types of discontinuity and the Intermediate Value Theorem • Cross-Lesson: 12.02 Power Series and Taylor Series

[The Fundamental Theorem · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 [Challenging] • Target: Continuity on an interval • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

[Limits & Continuity · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 [Challenging] • Target: Applying the Intermediate Value Theorem • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Level 3 · AdvancedLet $f(x) = x^3 - 3x - 1$. Explain how the Intermediate Value Theorem guarantees a root in $[1, 2]$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Testing continuity

A continuous function on a closed, bounded interval $[a, b]$ is guaranteed (Extreme Value Theorem) to be:

a. bounded and to attain its max and min

b. monotonic

c. unbounded

d. differentiable

22 [Foundational] • Target: Types of discontinuity and the Intermediate Value Theorem

Removable discontinuity: $f(x) = \frac{x^2-4}{x-2}$. To make f continuous at $x = 2$, define $f(2) =$

a. 4

b. 2

c. 0

d. -4

23 [Foundational] • Target: Continuity on an interval

By the Intermediate Value Theorem, does $f(x) = x^3 - x - 1$ have a root in $[1, 2]$?

a. No $--$ $f(1)$ and $f(2)$ have the same sign

b. No $--$ f is not continuous on $[1, 2]$

c. Yes $--$ $f(1)$ and $f(2)$ have opposite signs

d. Cannot be determined without more information

24 [Foundational] • Target: Applying the Intermediate Value Theorem

Does $f(x) = \frac{x-1}{x-1}$ have a removable discontinuity at $x = 1$?

a. No $--$ it is an infinite discontinuity

b. No $--$ f is continuous everywhere

c. No $--$ it is a jump discontinuity

d. Yes $--$ the factor cancels, leaving a hole

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25 [Foundational] • Target: Visualizing discontinuities

Intermediate Value Theorem guarantees a root of $f(x) = x^3 - x - 2$ on:

a. $(-1, 0)$

b. $(0, 1)$

c. $(1, 2)$

d. $(2, 3)$

26 [Foundational] • Target: Continuity of composite and combined functions

Global extrema on closed interval: Extreme Value Theorem guarantees max and min are attained. Find them by checking:

a. all critical points and endpoints

b. only critical points

c. only endpoints

d. only inflection points

27 [Foundational] • Target: Removable discontinuities and redefinition

If f has a jump discontinuity at $c \in (a, b)$, the accumulation function $F(x) = \int_a^x f(t)dt$ is still:

a. continuous at c (though not differentiable there)

b. discontinuous at c

c. differentiable at c

d. undefined at c

28 [Foundational] • Target: The Extreme Value Theorem (a continuity consequence)

Extreme Value Theorem: continuous f on $[a, b]$ attains:

a. only a maximum

b. only a minimum

c. both max and min

d. neither

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Testing continuity

If F and G are both antiderivatives of the same continuous f on an interval, then $F - G$ is:

a. a constant function

b. strictly increasing

c. equal to f

d. not necessarily differentiable

30 [Intermediate] • Target: Types of discontinuity and the Intermediate Value Theorem

If f is continuous on $[0, 1]$ with $f(x) \in [0, 1]$ for all x, the Brouwer Fixed-Point Theorem guarantees:

a. f is differentiable

b. $f(x) = 0$ for some x

c. $f(c) = c$ for some $c \in [0, 1]$

d. f is monotone

31 [Intermediate] • Target: Continuity on an interval

Find the hole (removable discontinuity) of $f(x) = \frac{x^2 + x - 2}{x^2 + 2x - 3}$.

a. x=-3

b. x=-2

c. x=1

d. no hole

32 [Intermediate] • Target: Applying the Intermediate Value Theorem

Classify the discontinuity of $f(x) = \frac{x^2 - 9}{x - 3}$ at $x = 3$.

a. jump discontinuity

b. infinite discontinuity

c. no discontinuity

d. removable discontinuity

33 [Intermediate] • Target: Visualizing discontinuities

A step function like the greatest-integer function $f(x) = \lfloor x \rfloor$ has ___ discontinuities at every integer.

a. infinite

b. removable

c. no

d. jump

34 [Intermediate] • Target: Continuity of composite and combined functions

For the Extreme Value Theorem to guarantee a global max and min, f must be continuous on:

a. all real numbers

b. an open interval

c. a single point

d. a closed interval

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35 [Intermediate] • Target: Removable discontinuities and redefinition

A function with a vertical asymptote at $x = 2$ has a(n) ___ discontinuity there.

a. removable

b. jump

c. infinite

d. oscillatory

36 [Intermediate] • Target: The Extreme Value Theorem (a continuity consequence)

If $f(x) = \frac{x^2 - 9}{x - 3}$, which type of discontinuity occurs at $x = 3$?

a. Jump

b. Infinite

c. Removable

d. Essential

37 [Intermediate] • Target: Testing continuity

If f has a jump discontinuity at $c \in (a, b)$, then $G = \int_a^x f\,dt$ at c is:

a. discontinuous

b. differentiable

c. not differentiable but continuous

d. zero

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Types of discontinuity and the Intermediate Value Theorem • Cross-Lesson: 12.02 Power Series and Taylor Series

Which function has a jump discontinuity at $x = 0$?

a. $f(x) = x^2$

b. $f(x) = 1/x$

c. $f(x) = |x|/x$

d. $f(x) = \sin x$

39 [Challenging] • Target: Continuity on an interval • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Change of variables $\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$ is a consequence of:

a. MVT

b. chain rule + FTC

c. IBP

d. Cauchy's integral formula

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40 [Challenging] • Target: Applying the Intermediate Value Theorem • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Mean Value Theorem guarantees c in open interval — this means the point is:

a. an endpoint

b. strictly interior

c. a maximum

d. undefined

41 [Challenging] • Target: Visualizing discontinuities • Cross-Lesson: 15.01 Double Integrals

Let f be continuous on $\mathbb{R}$ and $f(0) = 0$. If $f(x + y) = f(x)f(y)$ for all x, y, then:

a. $f(x) = 0$ everywhere

b. $f(x) = e^{cx}$ for some c

c. $f(x) = x$

d. $f(x)$ is periodic

42 [Challenging] • Target: Continuity of composite and combined functions • Cross-Lesson: 16.01 Vector Fields and Line Integrals

IBP formula follows from FTC because $\int_a^b fg'\,dx = [fg]_a^b - \int_a^b f'g\,dx$ derives from:

a. Differentiating fg and applying FTC Part 2

b. L'Hôpital

c. Green's theorem

d. Cauchy-Schwarz

43 [Challenging] • Target: Removable discontinuities and redefinition • Cross-Lesson: 2.01 Evaluating Limits

$\int_0^1 \frac{1}{x^p}\,dx$ (improper at 0) converges if and only if:

a. $p < 1$

b. $p > 1$

c. $p \leq 0$

d. never converges

44 [Challenging] • Target: The Extreme Value Theorem (a continuity consequence) • Cross-Lesson: 3.01 The Derivative from First Principles

Newton's law of cooling: $T(t) = T_e + (T_0 - T_e)e^{-kt}$. If room temp $T_e = 20$, initial $T_0 = 80$, $k = 0.1$, $T(10) =$

a. $20 + 60$

b. $60e^{-1}$

c. $80e^{-1}$

d. $20 + 60e^{-1} \approx 42.1$

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45 [Challenging] • Target: Testing continuity • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Let $F(x) = \int_0^x (t^2 + 5t + 4)\,dt$. At which x does F have a local maximum?

a. -1

b. $-\frac{5}{2}$

c. 0

d. -4

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